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Economic Analysis of Global and Local Policies for Respecting Planetary Boundaries

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ABSTRACT

In a series of highly cited papers over the period of 2009–2023, earth system scientists have identified a set of nine planetary boundaries that must not be breached if we wish to avoid catastrophic consequences for nature and humanity. These range from well-mixed, global boundaries, such as climate-altering atmospheric greenhouse gas concentrations, to localized limits on freshwater availability and reactive nitrogen entering the environment. Recent estimates suggest that six of the nine planetary boundaries have already been breached. The food system is a key driver of these exceedances and, therefore, must play a key role in any solutions. However, the establishment of these boundaries and the analysis of potential solutions have often been devoid of economic considerations. Furthermore, in the case of several of these planetary boundaries, limited attention has been given to the economic policies that might allow society to address them, as well as the likely synergies and tradeoffs across economic policies targeted to individual objectives. This paper seeks to bring further economic analysis to bear on the quantitative assessment of global and local economic policies aimed at respecting these planetary boundaries, concluding with seven lessons to inform future research on this topic.

JEL Classification: Q00, Q01

1 | Introduction

This paper focuses on the food system as a key driver of planetary boundary exceedances as well as a source of sustainability solutions. Planetary boundaries are absolute limits on human-induced environmental pollutants which, if exceeded, will cause irreversible damage to the earth systems upon which we rely. The most recent assessment of these boundaries quantified nine such boundaries, of which six have been exceeded (Richardson et al. 2023). Our food system currently accounts for one-third of greenhouse gas emissions (Crippa et al. 2021). It also results in land use change that is a key driver of tropical deforestation (Busch and Ferretti-Gallon 2017), which is, in turn, the main source of biodiversity loss on the planet (Leclère et al. 2020). Irrigated agriculture is also the most important user of freshwater and is a key driver behind groundwater depletion (Wada and Bierkens 2014). Farming is also the largest source of reactive nitrogen and phosphorous entering the environment (Gerten

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et al. 2020). Finally, agriculture is also an important source of novel entities introduced into the environment, including synthetic chemicals and pesticides (Persson et al. 2022). In short, the food system plays a significant role in the exceedances of all of the most problematic planetary boundaries identified in the most recent assessment. Furthermore, recent estimates suggest that developments in the global food system in the coming decades will only worsen these transgressions (Valin et al. 2021). In light of these heightened pressures, what can be done to lessen the adverse impacts of the food system on our environment?

While the consumption of locally produced foods is a popular concept, today's food system is increasingly globalized. Supply chains now span national and continental boundaries. Global change drivers, including income, population, biofuels, and technological change over the 2010–2050 period are projected to boost GHG emissions from the food system by more than one-third, increase nitrogen and phosphorous applications by as much as 50% and boost cropland and irrigation significantly as well (Valin et al. 2021). This translates into additional stress on the planetary boundaries in the absence of new and reformed policies.

Viewed through the lens of UN-FAO's recent State of Food and Agriculture report (Food and Agriculture Organisation of the United Nations 2023), the aggregate cost of these environmental damages is large—amounting to roughly one-third of global agricultural value-added. These costs are proportionately highest in the low-income countries. That same report also seeks to quantify the true cost of the food system on human health, as conveyed through dietary choices. These costs are even higher. However, the present paper will not address the dietary dimension of the global food system, focusing instead solely on the environmental impacts and the role of economic policies in mitigating these damages. Furthermore, most of the examples discussed here relate to the impacts of crop production. This is due to the reliance on gridded data and multiscale, global modeling of agriculture for which research on crop production is currently most advanced. However, livestock production is a major source of earth systems stresses (Qaim et al. 2024), and this deserves equal attention in future sustainability studies as data and modeling tools in this area are further advanced.

Unfortunately, most research to date on planetary boundaries has been policy-agnostic (Biermann and Kim 2020). This paper will explore the role of economic policies in mitigating the impact of global food systems on four of the key planetary boundaries: biodiversity loss, climate change, degradation of water quality, and groundwater depletion. It will highlight the importance of capturing the interplay between biophysical and economic systems, demonstrating how the omission of these feedbacks can lead scientists to overstate the costs of conservation while ignoring key benefits from respecting planetary boundaries. Market-mediated effects of such conservation policies play an important role in these outcomes, and the examples provided here highlight the need for greater engagement by economists in the debate over the future evolution of earth systems.

2 | Biodiversity and Ecosystem Services

Based on the most recent assessment of planetary boundaries (Richardson et al. 2023), it is clear that one of the planetary boundaries most severely compromised is genetic biodiversity. Here, irreversible losses are accumulating at a rapid pace (Ceballos et al. 2015). It is estimated that, of an estimated 8 million animal and plant species on earth, roughly 1 million are threatened with extinction (IPBES 2019), and roughly 10% of genetic diversity has been lost over the past 150 years (Exposito-Alonso et al. 2022). In a seminal paper titled "Bending the Curve of Terrestrial Biodiversity Needs an Integrated Strategy," Leclère et al. 2020) utilize a multi-model ensemble to highlight the crucial role of the food system and land use change in driving historical and future biodiversity loss. Under their business-as-usual scenario, they estimate that the world is currently on course to lose 13% of suitable habitat and 54% of wildlife density by 2100. However, they also identify a range of interventions that could contribute to reversing this trend or, in their terminology, "bending the curve" of biodiversity loss. These include supply-side measures involving sustainable intensification of farming and increased agricultural trade, as well as demand-side measures including reducing food waste and limiting animal product consumption. Finally, they explore conservation measures, including extending protected areas, conservation planning, and restoration of degraded habitats. When taken together, these ambitious changes to the food and natural resource systems could reverse the downward trend in biodiversity.

Unlike some of the other planetary stresses (e.g., greenhouse gas emissions), biodiversity richness tends to be highly concentrated (Jenkins et al. 2013), and the threatened vertebrate species reside in a highly localized subset of these regions. Jenkins et al. (2013) find that threatened birds are concentrated in the Andes and Southeast Brazil, along with Southeast Asian islands. Threatened mammals are most concentrated in Southeast Asia, while amphibians at risk of extinction are globally scattered but highly localized. In short, *geography matters* when it comes to potential biodiversity losses. This has spawned a number of studies seeking to link growth in population and demand around the world to biodiversity via global supply chains. Lenzen et al. (2012) link animal species threats to land use, commodity production, and international trade and argue that 30% of threatened species are linked to trade. Chaudhary and Kastner (2016) undertake a similar exercise and conclude that 83% of total species loss is owed to the conversion of natural lands to agriculture.

But how will these losses evolve in the future? Will the dramatic slowdown in population growth in many countries lessen this pressure for land conversion and biodiversity losses? Cisneros-Pineda et al. (2024) investigate this question using a global trade model (GTAP-AEZ) as modified for the study of biodiversity and ecosystem services (Johnson et al. 2023). They isolate the impact of population growth in each of 37 countries and regions of the world on land use change and biodiversity in every other region, measured as potential species loss. And this is done both for a historical period of demographic change (2001–2021) and a comparable future period (2021–2041). Pineda et al. (2024) find that the largest economic region in their model, the EU, will reverse the biodiversity impact it exerts on most other countries in the future due to their population decline. China also shows a similar reversal with its population decline alleviating pressure on global biodiversity. However, demographic trends in other regions lead to heightened biodiversity stress, most notably Nigeria, Ethiopia, and other countries in Sub-Saharan Africa

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(SSA). On net, despite the slowdown in the global population growth—and looming population declines in some of the richest nations—the future period does not show a declining rate of global biodiversity loss. However, the uncertainty associated with demographic change in the SSA region introduces enormous uncertainty into the estimates of future biodiversity losses.

While much of the literature on agriculture and biodiversity highlights the adverse impacts of the food system, there is another strand of research that focuses on the potential for biodiversity-agricultural production "mutualism" (Seppelt et al. 2020). This captures the idea that ecosystem services, such as natural pollination and sediment retention, can enhance agricultural productivity. However, these authors also make the point that ignoring this mutualism can lead to an adverse feedback loop wherein, for example, increased chemical use degrades the natural environment, leading to lower yields and, therefore, requiring additional cropland expansion, potentially leading to further intensification. The additional infringement on natural areas and associated ecosystem services can lead to further yield losses. The emerging field of agroecology aims to address this challenge by integrating the science of ecology with agricultural practice and societal values (Wezel et al. 2014). Agroecology seeks to maintain long-term productivity while also limiting adverse environmental impacts (Gliessman 2015). To date, agroecology has been predominantly the domain of niche operations (Ewert et al. 2023). However, this transdisciplinary, systems approach to studying agriculture is gaining significant traction—particularly in Europe—where it is reshaping the way farmers and policy makers think about farming. The current challenge lies in integrating agroecological research with research on the broader food system and finding ways to scale up these technologies so that they can contribute to the broader sustainability challenges (Ewert et al. 2023).

Closely related to the concept of biodiversity-production mutualism is the literature on more comprehensive measures of total factor productivity (TFP), often termed "Green TFP" or "Total Resource Productivity" (Fuglie et al. 2016; Ehui and Spencer 1993; Bureau and Antón 2022), which proposes to take agriculture-environment mutualism into account quantitatively by modifying TFP measures in such a way that pollution is treated as a negative output and common property resource extraction is counted as a costly input (Chambers 2016). This more complete accounting of the environmental costs of new technologies parallels the macroeconomic efforts to estimate "Green GDP." Johnson et al. (2023) pick up on these ideas and develop a framework for linking economic and ecological outcomes to assess the economic benefits of environmental conservation aimed at enhancing natural capital. Their GTAP-InVEST modeling framework allows for analysis of the evolution of six different land-based ecosystem services alongside growth in the global economy. These include: sediment retention, pollination of crops, coastal protection, climate regulation, timber production, and marine fisheries. The novelty in their analysis is that these ecosystem services, each measured using different ecosystem services models, evolve along their economic baseline (2021–2030 in this case) and feed back into the economic model via sector-specific productivity shocks. These authors find that, under a business-as-usual baseline estimated using the global economic model, the degradation of natural capital and associated ecosystem services, reduces global annual GDP by US $75 Billion in 2030. Importantly, the burden falls disproportionately on the world's poorest countries.

Johnson et al. (2023) also explore potential conservation policies aimed at offsetting the ecosystem services' degradation anticipated in their baseline simulation. They focus their attention on repurposing agricultural subsidies, investing the funds instead in nature-smart policies, including land rental payments for ecosystem services, and accelerated agricultural R&D. With these policies in place, they find that the GDP loss from degrading ecosystem services could be turned into a gain, ranging from US $25–$125 Billion in 2030. Furthermore, the gains would disproportionately benefit the lowest-income countries. These results echo the findings of Gautam et al. (2022), who provide a more comprehensive analysis of repurposing agricultural support to achieve a more sustainable food system. They find that simple reductions in, or rearrangement of, current support are unlikely to yield significant environmental benefits. Instead, these authors also argue for investing these funds in the development and adoption of green innovations, in effect allowing farmers to "do more with less."

The role of productivity in protecting the environment has been gaining increased attention (Coomes et al. 2019; Fuglie et al. 2022), and it has also been the subject of a series of OECD workshops. There is now substantial evidence that historical R&D investments aimed at generating new crop varieties have limited cropland expansion (Stevenson et al. 2012; Hertel et al. 2014; Villoria 2019). There is also emerging evidence that these benefits extend to limiting biodiversity loss—at least at the global scale. Baldos et al. (2025) assess the impact of the adoption of modern crop varieties (Fuglie and Echeverria 2024) on global croplands and biodiversity. While the global impact has been land-saving, the gridded analysis by Baldos et al. (2025) reveals that, at a local scale, the consequence of modern variety adoption has sometimes been cropland expansion. This outcome arises when technology adoption makes local producers relatively more competitive compared to producers elsewhere, leading to cropland expansion (Jevons' paradox). The impact of these new crop varieties on biodiversity depends critically on the gridded overlay of cropland change on these biodiversity hotspots. As with global cropland extent, the introduction of the CGIAR-related modern crop varieties (Fuglie and Echeverria 2024) has been globally beneficial to biodiversity. However, in the hotspots of cropland expansion, including in southeastern Africa and Southeast Asia, Baldos et al. (2025) find that the introduction of these modern varieties has resulted in increased potential species loss.

Rather than leveraging productivity growth to expand production, some high-income countries have taken advantage of TFP improvements to withdraw resources from agriculture in favor of the environment (Heisey and Fuglie 2018). Nowhere is this more evident than in Northwest Europe, where there is a strong emphasis on agri-environmental policies (Wuepper et al. 2024). Restrictions on fertilizer and chemical use, as well as limits on animal numbers and cropland extent, along with reforms to the Common Agricultural Policy, have resulted in a secular decline in input usage in agriculture (Figure 1a) (Heisey and Fuglie 2018). Nonetheless, strong TFP growth has permitted this region to maintain agricultural output over this entire period (Figure 1a).

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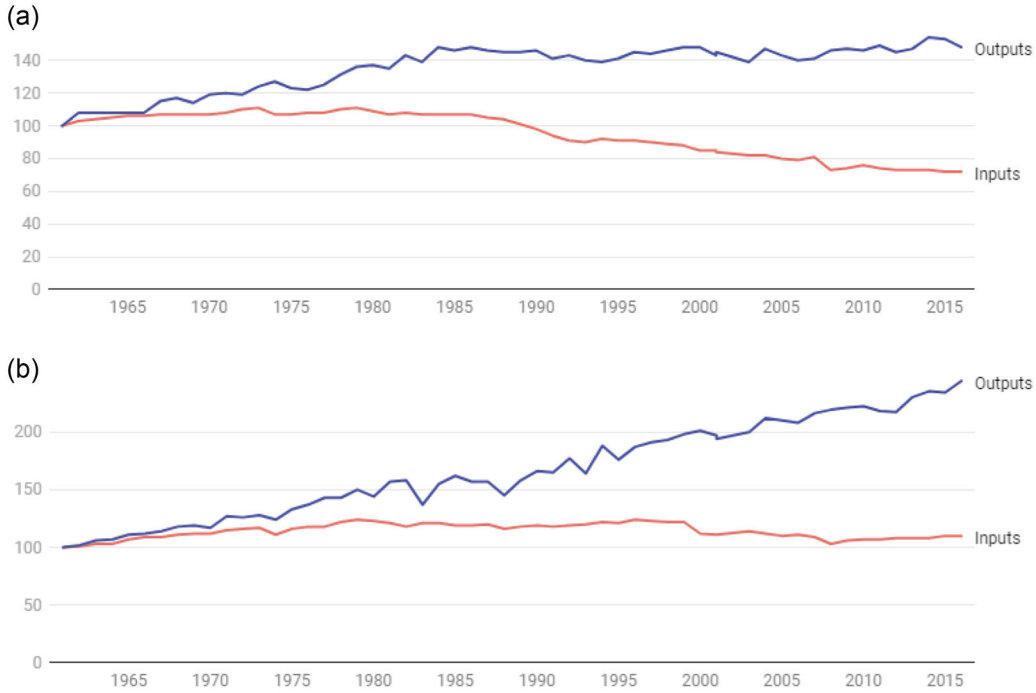

FIGURE 1 | Top (a) shows input and output growth in Northern Europe. The difference between the two lines is attributable to Total Factor Productivity (TFP) growth. (b) shows a comparable time series for North America. *Source:* Data taken from Heisey and Fuglie (2018) as presented in Hertel (2021).

This contrasts sharply with the pattern of input usage in US agriculture (Figure 1b), where the index of aggregate inputs has remained virtually unchanged over the past 50 years. Instead, reduced land and labor have been offset by increased spending on machinery and intermediate inputs such as seed and chemicals (Heisey and Fuglie 2018). As a result, strong TFP growth has been the dominant explanatory variable behind continuing farm output growth in the US. This implicit choice of 'how to use TFP growth' is largely a political decision—weighing the tradeoff between agricultural and environmental interests. In the EU, this tradeoff has clearly tilted in favor of the environment as reflected in the large number and intensity of environmental policies directed at the farm sector (Wuepper et al. 2024).

There are strengthening calls, worldwide, to set aside more land for nature (IPBES Secretariat 2019). Dubbed "30 by 30," the broad thrust of the most recent IPBES proposal is to set aside 30% of the earth's land and oceans for nature, by the year 2030 (Dinerstein et al. 2019). The precise details of where and how this would be achieved remain to be worked out, but given the ambition of these proposals, they will clearly have an impact on agricultural land use. In earlier work, Mehrabi et al. (2018) estimated that setting aside half the earth's surface for nature would reduce cropland area by 15%–31%, pastureland by 23%–25%, and crop calorie availability by 3%–29%. However, in addition to exploring a more ambitious target, their analysis did not factor in economic responses to the measures. Above all, most analyses to date have not considered how such a policy would be implemented. Would the land simply be set aside as a national or international park? Or would the policy be voluntary, with market incentives to set aside biodiversity-rich lands in sensitive regions? Would the market incentives involve taxes on the conversion of natural lands, or would it involve subsidies to restore commercial lands to their natural state? Would such subsidies vary with the agricultural productivity of the land being set-aside? Will there be a priority placed on agriculture-biodiversity mutualism? What metric for biodiversity would be employed? Would the policy be implemented at the local, regional, or national scale? Given the local nature of biodiversity hotspots, how would the effectiveness of the policy be reduced if the policy is implemented at the national, as opposed to local, scale? There are many practical questions that have not yet been addressed. Yet these have important implications for the impact which pursuing the 30×30 objective will have on the food system.

In ongoing research on this topic, Haqiqi et al. (2024) explore the potential economic impact of a global 30×30 policy which seeks to set aside biodiverse lands currently under cultivation, and protect biodiverse lands that are under natural cover. Using a spatially resolved land allocation model, they estimate that conserving 10 million km^2 of land would cost between \$71 and \$120 billion USD. This translates to an average cost of \$71 to \$122 per hectare, with significant spatial variability due to differing opportunity costs of land across locations. In China and Japan, conserving land may require up to \$500 per ha for highly profitable cropland. In Brazil and Sub-Saharan Africa, this can be as low as \$27 per ha. They find that the impact of this policy on global food production is surprisingly small (~ -2%) due to market adjustments and spillover effects.

Barbier and Burgess (2019) make the argument that establishing such absolute limits on land use is a necessary condition, but it is not sufficient for achieving the stated goal of preserving biodiversity and keeping the world within a safe operating zone. The problem is that the *absolute scarcity* of land induced by such policies will raise land rents and make the ultimate goal

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more difficult to achieve. They argue that landowners will be encouraged to convert all of those areas outside this narrow limit, thereby accelerating the loss of ecosystem services. Instead, these authors argue that the core of the problem rests with the undervaluation of, and underinvestment in, natural ecosystems. They argue for *managing relative scarcity* by boosting the cost of eroding ecosystem services, thereby sending economic signals to decision-makers to make better use of this natural capital. They provide an illustrative example of this challenge for the case of natural forests. Here, they conclude that cutting in half the rate of tropical deforestation without changing the economic policy environment would only extend the lifetime of the remaining safe operating space for these forests from 11 to 21 years. In contrast, an optimal economic policy aimed at raising the relative scarcity of these forests would treat the remaining safe operating space as a depletable resource which should be managed via a time varying tax on deforestation. They find that such an economics-based policy would greatly reduce the rate of depletion of the remaining safe operating space and extend the lifetime of this asset—potentially for as much as 200 years.

In a more in-depth case study of ecosystem protection, Barbier and Burgess (2024) explore these issues in the context of a case study of the attempts to preserve the world's peatlands. Peatlands comprise plant materials that have been partially decayed and accumulated over a long time period. While peatlands cover just 3% of the terrestrial planet (Humpenöder et al. 2020), they have been estimated to store twice as much carbon as all of the world's forests (Yu et al. 2010). Peatlands also provide a wide range of other ecosystem services ranging from the provision of unique habitats, supporting water cycles and filtering pollutants. Yet, despite these many contributions, peatlands are being rapidly drained and degraded. Barbier and Burgess (2024) point to the current mismanagement of these peatlands, arguing that they are fundamentally undervalued, with subsidies to agriculture, forestry, and mining often resulting in accelerated drainage and degradation of this natural resource. They recommend that countries with significant peatland areas should ensure that the services provided by these natural ecosystems are taken into account by those making land use decisions. Their second recommendation involves boosting technical and financial assistance for countries seeking to invest in their peatlands—not only protecting existing peatlands, but also "rewetting" drained peatlands so they can be restored to their natural state. While costly, this can be an important part of a global climate mitigation strategy (Humpenöder et al. 2020), a topic to which we now turn.

3 | Climate Change

When the term planetary boundary is mentioned, most scientists, as well as members of the public, are inclined to immediately think about climate change and the seemingly inexorable rise in greenhouse gas concentrations in the earth's atmosphere. In addition to the steady increases in global mean temperature, climate scientists have made great progress in the field of event attribution, linking some of the extreme events involving flooding, drought, and severe storms to the rise in GHG concentrations in the atmosphere (Otto 2017). These concentrations have risen from about 277 ppm in 1750 to 420 ppm today (US Department of Commerce). Since CO2 is a well-mixed, global pollutant, it does not matter where the GHGs are emitted, they have an impact across the entire planet. This creates unique opportunities from a mitigation policy perspective because it does not matter where the reduction in GHG emissions occurs. Therefore, the optimal global policy involves starting with the lowest cost mitigation options, worldwide. Of course, since many of these low-cost options are in the world's low-income countries, the question of who pays for the abatement is critical.

The idea of starting with the least cost GHG abatement is important for the food system, since the land-based abatement policies are often amongst the least cost options—specifically measures related to agriculture and forestry (Murray et al. 2005; Golub et al. 2009). McCarl and Schneider (2001) identify three different avenues through which agriculture can contribute to mitigating climate change: (1) direct reduction in emissions (e.g., through manure management, modified fertilizer use), (2) soil carbon sequestration and avoided deforestation, and (3) through production of commodities that substitute for fossil fuels (e.g., biofuels). Virtually all of these mitigation policies lead to increased food prices which can have adverse consequences for the poor who spend a disproportionate share of their income on food (Golub et al. 2012; Hasegawa et al. 2018; Havlík et al. 2015) and these low income households benefit little from the expected scarcity-induced increases in land returns (Hussein et al. 2013). When implemented at a global scale, a carbon pricing scheme would also penalize agricultural activities in low-income countries, which are typically characterized by very high emissions per dollar of output (Avetisyan et al. 2011). In practice, finding a policy combination that takes advantage of low-cost mitigation options, while also avoiding deleterious effects in the poorest countries and households is a challenging task (Henderson et al. 2017). Furthermore, absent the capability of actually monitoring emissions from millions of farms to ensure that less emissions-intensive practices are put in place, a carbon tax must essentially resort to reducing output of the polluting activities, which in turn raises food prices (Martin and Vos 2024).

An alternative to farm-level regulation, taxes or other cost-increasing measures is to make public investments in new technologies that have the potential to increase output per unit input and therefore lower costs while also reducing GHG emissions for a given supply of food. These investments generate knowledge capital, which raises productivity, not only in the country making the investment, but often in other countries—through knowledge spillovers (Fuglie, 2018). There is a long tradition of studying the relationship between R&D and productivity in agricultural economics and the evidence is that these investments yield a high economic return (Alston et al. 2010). However, less attention has been given to the environmental impacts of such investments. Fuglie et al. (2022) argue that GHG emissions from agriculture are closely tied to total input use, and since historical investments in productivity-enhancing public R&D have limited the growth in agricultural inputs, they have also served to limit GHG emissions from agriculture. In considering a range of future investment scenarios, they argue that accelerated R&D investments could achieve abatement at a cost as low as $11/ton CO2e if combined with policies preventing agricultural conversion of carbon-rich natural lands. This compares very favorably with other mitigation options. Furthermore, unlike on-farm mitigation measures, the public R&D investment policy does not require regulatory com-

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pliance by hundreds of millions of individual farmers. Fuglie et al. (2024) provide a comprehensive review of agricultural productivity and its consequences for climate mitigation and conclude that this avenue offers relatively low-cost mitigation (less than $50/ton CO2e). Furthermore, by targeting such investments to the more emissions-intensive commodities in agriculture, they argue that marginal abatement costs could be far lower than this estimate. This point is further developed by Gautam et al. (2022), who argue that green innovations should be specifically focused on methane-intensive production activities (rice and ruminants) where there appear to be significant opportunities for reducing emissions of the highly potent GHG while also improving productivity.

An additional benefit of the productivity-enhancement approach to GHG mitigation is that, when employed broadly, it lowers the cost of complementary policies aimed at removing ecosystem-sensitive lands from production. By boosting output on existing agricultural lands, commodity prices are lowered, and land rents are also reduced, relative to baseline. Therefore, the subsidy required to prevent the conversion of natural lands is also reduced. Fuglie et al. (2022) find that this policy synergy, between enhanced public R&D in agriculture on the one hand, and the protection of carbon-rich natural lands on the other, results in a 10% reduction in the cost of implementing global land conservation. Martin and Vos (2024) also point out another benefit of the R&D-led approach to GHG mitigation: reduced leakage from country-specific investments as other countries are confronted with lower commodity prices as a result of the innovating region's production increase. This contrasts with a regulatory approach to limiting emissions, which raises prices and encourages production increases in more polluting regions.

There are additional co-benefits from climate mitigation policies—particularly reductions in fossil fuel combustion where significant health and environmental gains have been identified (Ürge-Vorsatz et al. 2014). However, apart from the extensive literature covering climate impacts and mitigation in agriculture, the interplay of climate policy, food systems, and other planetary boundaries has been less fully explored. In a recent study of climate policy in the United States, Zuidema et al. (2023) uncover important water quality benefits from a carbon-based pricing scheme (Figure 2). The mechanism underlying this linkage is as follows. Carbon pricing raises the cost of carbon-intensive goods, with the cost rising roughly in proportion to the cost share devoted to carbon-intensive inputs. Very few commodities are as carbon intensive as nitrogen fertilizer, where the cost share of natural gas in anhydrous ammonia (which serves both as a feedstock and as an energy source for the synthesis reaction in the Haber-Bosch process) is very significant. When evaluated at a social cost of carbon of $176/ton CO2e—the high end of US estimates (U.S. Interagency Working Group 2021), Zuidema et al. (2023) estimate that natural gas costs would rise by 176% and anhydrous ammonia prices would nearly double. This, in turn, has a significant impact on nitrogen fertilizer applications in the US Corn Belt. Zuidema et al. (2023) predict that nitrogen fertilizer use would fall by about 14%—half due to a reduction in corn production, and half due to reductions in application rates (Figure 2). With less nitrogen fertilizer being applied, there is less nitrate leaching. The authors estimate that this translates into about 8% less nitrate pollution going into the groundwater and with a similar reduction in nitrate export to the Gulf of Mexico. The smaller-than-proportional decline in nitrate exports stems from the natural ecological processes that occur between the runoff from a farmer's field and the ultimate delivery to the Gulf (Zuidema et al. 2024). Given the lack of success to date with voluntary programs aimed at reducing non-point source pollution from US agriculture, this is a significant impact. Indeed, Zuidema et al. (2023) estimate this reduction in nitrate exports to the Gulf of Mexico to be comparable to, or even greater than, the gains offered by widely advocated wetlands restoration programs. In sum, these water quality improvements represent an important, positive synergy between the climate and biogeochemical planetary boundaries.

4 | Degradation of Water Quality

While there are potential synergies between climate policy and water quality, when viewed through the lens of planetary boundaries, it is the biogeochemical flows—in particular nitrogen (N) and phosphorous (P)—that present the more severe exceedances (Richardson et al. 2023). Furthermore, unlike climate change, these are not well-mixed, global pollutants, but rather they are the result of complex processes—often quite localized—that are not always well understood or measured but which can have widespread impacts on the environment and human health (Vitousek et al. 1997; Galloway et al. 2003). There is a rich history of attempts to deal with N and P pollution at a local level through a variety of policies, including more efficient use of nutrients in-field, as well as edge-of-field mitigation policies (Iowa State University 2013). The most cost-effective policy will inevitably depend on a whole host of considerations (Christianson et al. 2013). A recent study by Liu et al. (2023) looks across the entire US Corn Belt, considering four different types of policies, including taxation tied to nitrate leaching rates, improvements in farm practices to increase nitrogen use efficiency including split applications, as well as controlled drainage of subsurface tile systems, and wetland restoration. Those authors find that the most effective policy varies widely across the Corn Belt (Figure 3), suggesting the value of locally targeted policies.

Zuidema et al. (2023) highlight the potential spillover effects from policies that only apply in certain grid cells, but not in others. Consider, for example, the wetland restoration policy—, which can only be implemented in locations suitable for wetlands. By removing farmland from production, this policy raises crop prices, while also lowering overall fertilizer use and prices. This encourages additional production and fertilizer applications elsewhere in the region (Figure 4, panel a). However, by pairing the wetlands policy with a broad-based policy, such as the carbon-pricing scheme discussed previously, these spillover effects are largely eliminated (Figure 4, panels b and c).

These findings highlight the fact that, while water quality is a local problem, addressing it can have broader impacts—often conveyed through market mechanisms. This is particularly true when one considers potential solutions at a global scale. Gerten et al. (2020) have suggested that a reallocation of nitrogen fertilizers globally could allow producers to feed the world adequately while respecting multiple planetary boundaries. Smerald et al. (2023) reach a similar conclusion, arguing that fertilizer use could be reduced by one-third, while maintaining the global production

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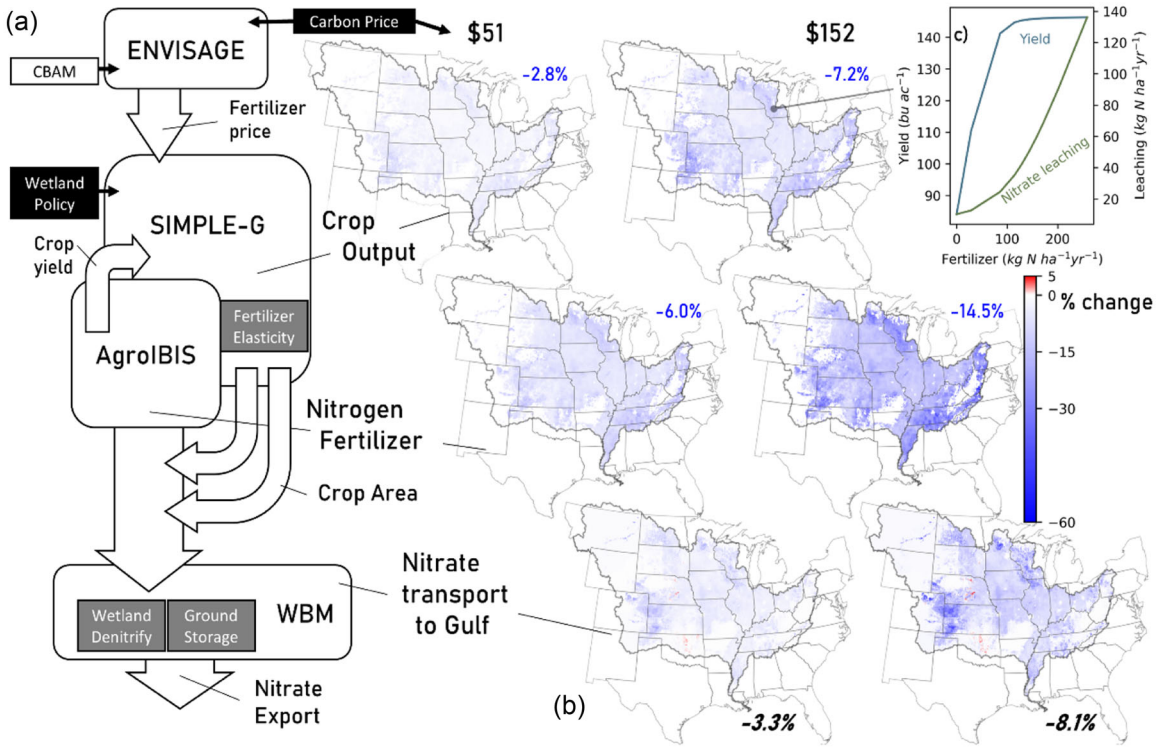

FIGURE 2 | Cascading effects of US climate mitigation through carbon pricing at $51 and $152/tonne of CO2. Costly natural gas raises the price of nitrogen fertilizer and shifts the demand and supply curves for crop production (EVISAGE model). These effects are incorporated into the gridded SIMPLE-G model of corn and soy production, reducing nitrate leaching (Agro-IBIS model) and hence lowering pollution of ground and surface water. Discharge into the Gulf of Mexico falls by 8.1% at the higher carbon price (Water Balance Model estimates). *Source:* Zuidema et al. (2023).

of major cereals, simply by reallocating the global distribution of fertilizers. However, these authors do not propose specific policies to achieve this reallocation. Furthermore, they do not explore the potential economic impacts of achieving this goal. What would be the consequences for crop prices, employment and trade? Any attempt to sharply limit fertilizer use in some locations will have important consequences for international trade, both in the reallocation of nitrogen-intensive crops and in the reallocation of nitrogen fertilizer. This is a topic that deserves further investigation by trade economists, in collaboration with biophysical scientists.

5 | Groundwater Depletion

Freshwater availability is a planetary boundary that has begun to receive more attention from earth system scientists (Richardson et al. 2023). Water shortages are of critical significance to the global food system, as agriculture is the largest consumer of freshwater worldwide, accounting for roughly 70% of all freshwater withdrawals, of which more than half is 'consumed' in the form of evaporation and transpiration (Wisser et al. 2010). Groundwater withdrawals are a growing share of total freshwater usage. Wada et al. (2014) estimate that global groundwater depletion tripled over the 50 years from 1960 to 2010 and they project that it will double again by the end of this century, resulting in 40% of human water consumption being supplied by from nonsustainable resources by 2100 (Wada and Bierkens 2014). This has led to dire predictions about global food security as groundwater supplies are exhausted (Famiglietti 2014).

In a recent paper, Haqiqi et al. (2022) explore the connection between groundwater, irrigation, agricultural production, and food security using a gridded economic modeling framework: SIMPLE-G (Haqiqi and Hertel 2024). Specifically, the authors assess the impacts of a global groundwater policy that restricts excessive groundwater withdrawals to their estimated rate of average annual recharge. This policy has spillover effects for land use, as those areas currently exceeding planetary boundaries, via excessive withdrawals, account for more than 12% of global crop production (Figure 5, first bar). If there were no further adjustments, and all of the crop production relying on unsustainable groundwater withdrawals were lost, Famiglietti's (2014) prediction of "skyrocketing food prices and profound economic and political ramifications" would indeed be realized. Assuming a global price elasticity of demand for food of 0.33 and, therefore, a price flexibility of 3, this would result in a permanent 36% rise in global crop prices, with much larger increases in the most affected locations. This would have a disastrous impact on food security.

However, when prices rise, there are a variety of market-mediated responses, starting with the substitution of surface water for groundwater, where this is feasible (Figure 5). Deficit irrigation (reducing the frequency and/or amount of irrigation water applied) as well as investments in improved irrigation equipment (e.g., drip irrigation) to make better use of scarce water are additional on-farm adaptations that are expected in the face of scarcity. Taken together, Haqiqi et al. (2022) estimate that these on-farm measures could offset about one-quarter of the production loss (3.5%) at the global level (Figure 5, second bar). A second avenue for adaptation is to stop irrigating altogether,

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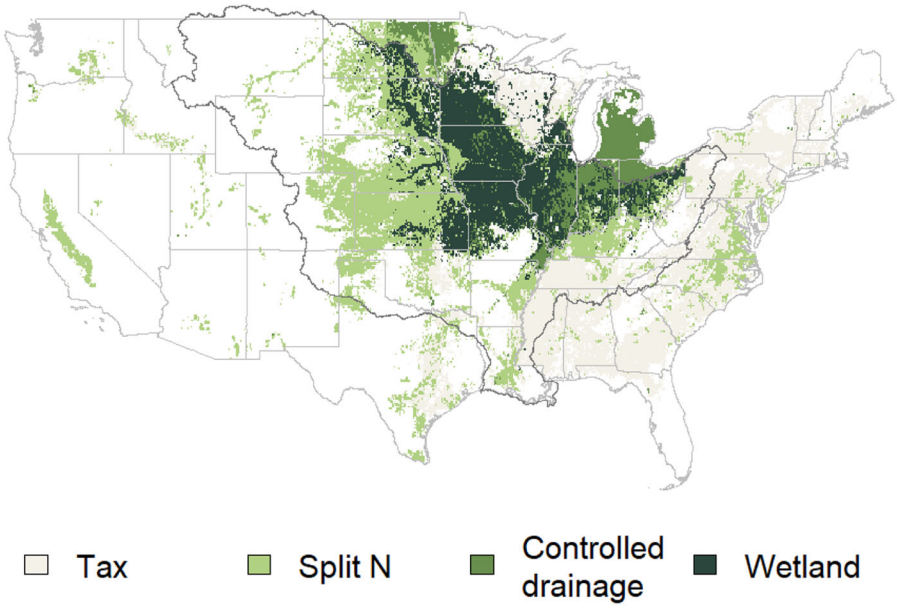

FIGURE 3 | Most effective mitigation policy for mitigation of nitrate pollution in the US Corn Belt, by grid cell. Tax, refers to the taxation of leaching, Split N refers to multiple applications of nitrogen fertilizer to increase nitrogen use efficiency. Controlled drainage refers to the use of an edge of field instrument that can be used to limit runoff from subsurface drainage systems during periods when in-field activities are not required. Finally, Wetland refers to the restoration of a wetland at the edge of a field to capture runoff and facilitate denitrification. *Source:* Liu et al. (2023).

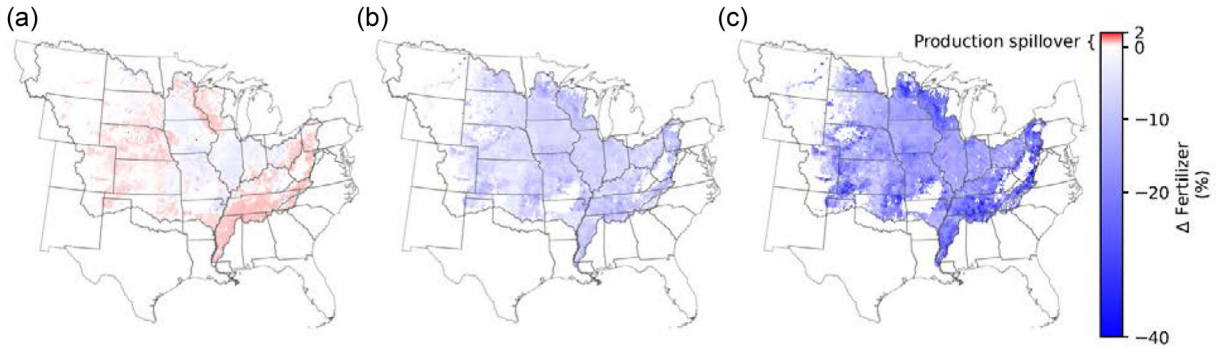

FIGURE 4 | Effect on nitrogen fertilizer applications Effect on nitrogen fertilizer applications of coupling wetland restoration with carbon pricing. (a) Absent carbon pricing ($0), restoration of wetlands over the tile-drained land causes market-mediated production spillovers of cropland beyond the central Corn Belt, thereby increasing local fertilizer use across much of the Mississippi Basin and offsetting the impacts of the wetland restoration policy. Raising fertilizer prices via carbon pricing nearly eliminates this spillover effect and decreases fertilizer applications over most of the Mississippi Basin at (b) $51 and (c) $152 carbon prices.

and allow the irrigated cropland to revert to rainfed production. The authors estimate that this would recoup another one-quarter of the total production loss (3.18%—third bar in Figure 5). A final adaption involves relocating production. In response to higher prices, locations that are not currently cultivated could be converted to cropland, thereby offsetting another 5.4% of the global production loss. This will involve both domestic relocation as well as increased food imports from water-abundant regions. Haqiqi et al. (2022) estimate that this increase in the "virtual water trade" could additionally offset more than 3% of the global production loss. In the end, they argue that the decline in global crop output would be less than 1% of total production (final bar in Figure 5). Of course, the authors point out, some of these adaptations will have a deleterious impact on other planetary boundaries as a consequence of cropland expansion into natural areas as well as increasing intensification of production in response to higher crop prices, and these must also be considered in a comprehensive analysis.

While the long-run impact of groundwater restrictions on global agricultural markets is unlikely to be as dramatic as anticipated by the physical scientists, these market-mediated responses can be devastating to local communities. Indeed, in many locations, vibrant agricultural economies have grown up around these groundwater-driven farming systems. What will happen to them if such sustainability policies are put in place? Haqiqi et al. (2022) identify local communities where agricultural employment might fall by as much as 50% in the face of these global groundwater restrictions. What are the implications of such dramatic shocks to the local economy? Ray et al. (2023) explore this question

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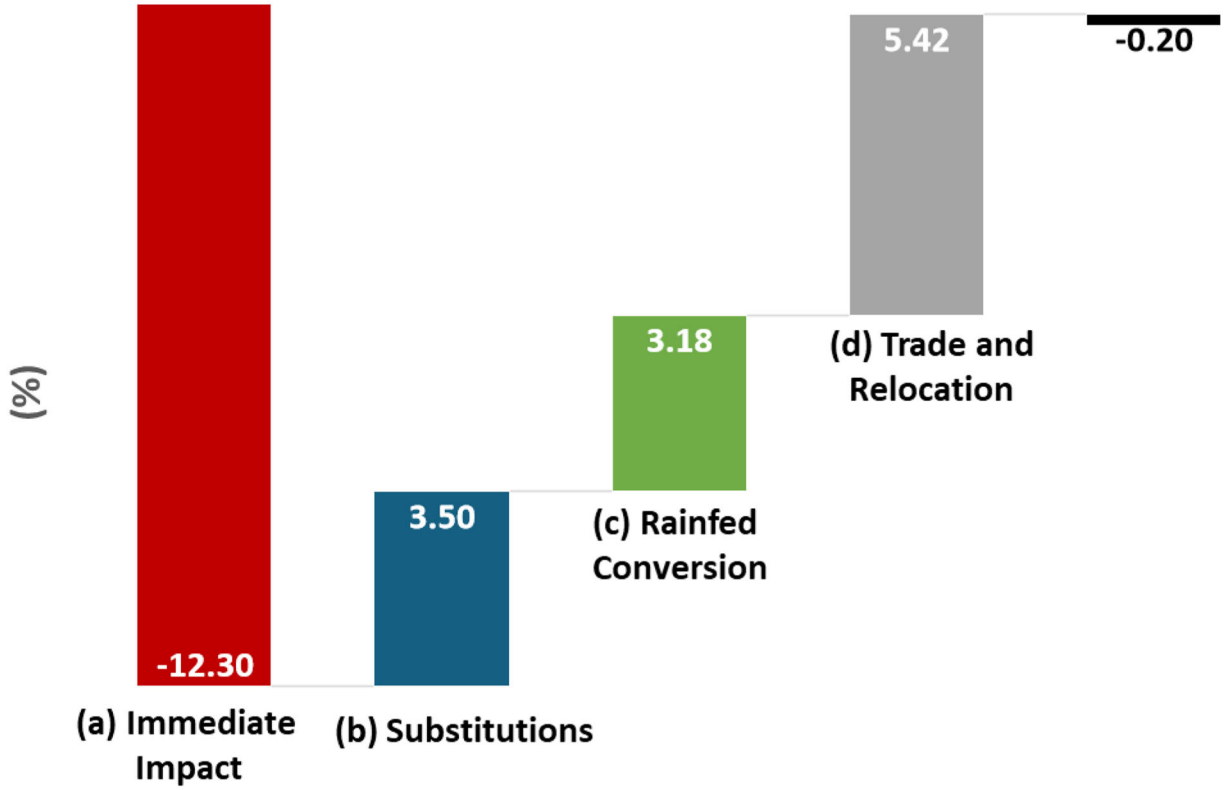

FIGURE 5 | Response of global crop output to global groundwater sustainability restrictions. Entry (a) is the impact of absent economic adaptations. Margins of adaptation include: (b) on-farm substitutions of other inputs for groundwater, including more efficient irrigation systems; (c) reversion to rainfed production; (d) relocation within the country and international trade. The final entry shows the impact of the sustainability policy on global crop output after economic adaptations have taken place. *Source:* Adapted from Haqiqi et al. (2022).

for the more limited case of groundwater restrictions in the Western US. They highlight the importance of labor market rigidities in governing the effectiveness and equity impacts of such conservation policies. In the US, there is recent evidence that farm labor is becoming less mobile (Fan et al. 2015). Ray et al. (2023) show that, as labor becomes less mobile, conservation policies tend to become less effective, even as these same policies have more dramatic impacts on local wages. Therefore, we cannot accurately assess the consequences of efforts to respect planetary boundaries without considering the potential (or lack thereof) for local adaptation and the consequences for household well-being.

6 | Summary and Conclusions: Seven Lessons From Economic Analysis of Planetary Boundaries

What lessons can be drawn from this literature and the associated attempts to bring economic analysis to bear on attempts to restore planetary boundaries associated with the food system?

Most research on planetary boundaries ignores economics and is policy agnostic: In a thoughtful review of the evolution of research on planetary boundaries, Bierman and Kim (2020) highlight the "boundaries of planetary boundaries." They note the apolitical, earth-system science-driven nature of the original planetary boundaries formulation led by Johan Rockström et al. (2009). Those authors did not attempt to include policy-makers and other stakeholders in these early discussions. Yet, as we have seen from the examples explored above, the economic policies required to protect these planetary boundaries can have dramatic impacts on local communities as well as entire countries. Furthermore, once one moves into the realm of economic policies, tradeoffs and synergies emerge that are often ignored by the earth system scientists who simply wish to make the food system "fit" into these planetary boundaries. Where rigid boundaries are prescribed, Barbier and Burgess (2019) demonstrate that, in the absence of appropriate pricing and investment in natural capital, these policies may actually speed up the demise of the remaining safe operating space. Of course, many of the proposed economic policies likely face stiff political economy opposition. Indeed, any transformation of the food system faces the challenge of confronting a broad range of polarizing factors that make such reforms particularly challenging (Resnick and Swinnen 2024). Nonetheless, there is great scope for economists to engage with, and make constructive contributions to, the community of scholars studying planetary boundaries.

Doing more with less (boosting TFP) offers a broad-based sustainability strategy which is complementary to conservation policies: As demonstrated in Fuglie et al. (2022), total factor productivity growth in agriculture over recent decades has allowed agricultural output to grow, even as composite input use has leveled off—or even declined—as has been the case in Northern Europe (Figure 1). Since much of the environmental damage from farming is associated with input use, whether it is cropland expansion, fertilizer and chemical usage, fuel combustion, or other factors,

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curtailing input use in agriculture has had beneficial consequences for the environment. Many of the improvements in TFP can be directly linked to past public investments in research and development (Uris Lantz C. Baldos et al. 2019; Julian M. Alston et al. 2010). This means that such investments can often be justified on environmental grounds. Indeed, Fuglie et al. (2024) argue that publicly funded agricultural R&D can be justified as a GHG mitigation strategy for agriculture, solely on the basis of the relatively low marginal cost of emissions abatement. Add to this the beneficial food security impacts of lower food prices, and it becomes a very attractive policy indeed. Furthermore, as Fuglie et al. (2024) show, this R&D strategy also lessens the cost of complementary land set-aside policies, by lowering the land rental payments required to remove cropland from production. Finally, as shown in Johnson et al. (2023), conserving cropland can have further benefits for agricultural productivity by promoting biodiversity-agriculture mutualism. In short, efforts to improve agricultural TFP offer a broad-based sustainability strategy.

Market-mediated spillovers are pervasive and give rise to tradeoffs across planetary boundaries: In failing to model the explicit policy interventions required to respect the planetary boundaries, earth system scientists typically overlook the way in which markets will respond to the desired interventions. For example, Gerten et al. (2020) demonstrate that "feeding ten billion people is possible within four terrestrial planetary boundaries" by reallocating agricultural production, water, and nitrogen use, while also reducing food loss and altering diets. However, this biophysical feasibility assessment does not address how these changes will be attained in a world of hundreds of millions of independent farmers. By embedding these biophysical relationships into an economic model as has been done by Liu et al. (2023) for the US, it will be possible to incorporate explicit policies as well as exploring the market-mediated spillover effects. These will include: the relocation of production, both within and across countries, as well as the reallocation of nitrogen fertilizer from targeted areas to other areas of agricultural production. Absent complementary policies, these market-mediated spillovers may exacerbate pollution elsewhere (Figure 4). *Synergies arise when conservation aimed at one planetary boundary reduces the intensity of production, thereby benefitting other planetary boundaries*: Intensive agriculture often places stress on multiple planetary boundaries. Nowhere is this more pronounced than in Northern India, where Green Revolution technologies, relying on intensive irrigation and fertilization of crops, coupled with government subsidies for groundwater pumping, have led to serious depletion of groundwater as well as water quality and health challenges (Brainerd and Menon 2014; Dangar et al. 2021). As shown by Haqiqi et al. (2022), policies focusing on reductions in groundwater withdrawals can offer a dual benefit of reducing fertilizer use through the relocation of production to areas where production is less fertilizer-intensive.

Market adaptations generally moderate the impact of planetary boundary conservation measures on food prices: A legitimate concern with enforcing planetary boundaries related to the food system is the resulting impact on food prices, and hence exacerbating food insecurity. The specter of food shortages arising from excessive groundwater depletion offers a case in point (Famiglietti 2014; Wada et al. 2012). However, Liu et al. (2014) demonstrate that international trade can play a significant role in moderating the impacts of future water scarcity on food security. Similar findings are obtained in the context of scenarios where groundwater depletion is curtailed to remain within planetary boundaries (recall Figure 5 above). In short, by ignoring economic adaptations, physical scientists tend to overstate the impact on food prices of localized interventions focused on conservation measures related to land and water use.

The impact of planetary boundary policies on local communities can be dramatic, and the community responses will shape the effectiveness and equity impacts of these policies: While economic adaptations can moderate the food price impacts of policies aimed at respecting planetary boundaries, they are likely to exacerbate the consequences for local communities. Consider parts of California's Central Valley, where local farming systems can rely heavily on unsustainable extraction of groundwater. In some cases, the enforcement of a sustainability standard will require the termination of groundwater pumping with dramatic consequences for local employment. Ray et al. (2023) estimate that wages in some parts of the Central Valley could fall by as much as 25%. Those authors show that the mobility of workers across sectors as well as across labor markets will determine not only the incidence of such a groundwater sustainability policy, but also the effectiveness of the policy itself in determining the stated goal. In short, we cannot ignore community responses in assessing the likely impacts of attempts to enforce planetary boundaries. This requires analysis far beyond the realm of the earth system scientists defining these environmental limits. In an attempt to address these issues of equity, the field of "Donut Economics" has emerged (Raworth 2017). Here, the idea is to treat the planetary boundaries as the outer layer of the donut, with the inner layer relating to the social foundation for a "regenerative and distributive economy." In response to these concerns, Rockström and colleagues have revised their concept of planetary boundaries—to focus on "safe and just earth system boundaries" in which social justice considerations further constrain these boundaries (Rockström et al. 2023).

Global-Local-Global analysis is essential for tackling planetary boundary challenges and identifying solutions: The strong temptation for many researchers studying planetary boundaries—particularly those related to land and water use—is to undertake location-specific studies. Indeed, as the agricultural economics profession is well aware, most sustainability issues are inherently local, with impacts and solutions depending on weather, soils, topography, technology, and community governance. However, the global food system is fundamentally demand-driven. Farmers grow crops in order to satisfy consumer demand, and, as the growth in global food system trade and investment will attest, potential consumers reside across the entire globe. In order to understand the drivers behind contemporary food system sustainability challenges, a global approach is required. Indeed, in a study of future groundwater stress in the Western US, Haqiqi et al. (2023) conclude that the most important drivers of unsustainable withdrawals in that region arise from population and income growth outside the United States. Local decision makers need access to estimates of these global change drivers in order to formulate effective policies. These local decisions will likely have spillover effects. Hence the need for global-to-local-to-global analysis of sustainability policies (Hertel et al.

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2023). This will require a synthesis of research focused on global economic analysis with studies undertaking geospatial analysis at the subnational level (Hertel et al. 2019). Recent efforts to develop multiscale frameworks that integrate local agricultural activity into regional, national and global models of the food system are promising (Valin et al. 2013; Lotze-Campen et al. 2008; Haqiqi and Hertel 2024). By modeling global agricultural activity at the level of individual grid cells, this research is more amenable to close collaboration with earth system scientists as well as other disciplines. Stimulating such collaborations in the context of global-to-local-to-global analysis is the goal of the NSF-funded GLASSNET project (https://glassnet.net). Readers are invited to join this network of networks and participate in the associated workshops, early career scholar exchanges and training courses.

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References

Alston, J. M., J. S. James, M. A. Andersen, and P. G. Pardey. 2010. *Persistence Pays: U.S. Agricultural Productivity Growth and the Benefits From Public R&D Spending*. Natural Resource Management and Policy 34. Springer New York. http://link.springer.com/chapter/10.1007/978-1-4419-0658-8_8.

Avetisyan, M., A. Golub, T. Hertel, S. Rose, and B. Henderson. 2011. "Why a Global Carbon Policy Could Have a Dramatic Impact on the Pattern of the Worldwide Livestock Production." *Applied Economic Perspectives and Policy* 33, no. 4: 584–605. https://doi.org/10.1093/aepp/ppr026.

Baldos, U. L. C., F. G. Viens, T. W. Hertel, and K. O. Fuglie. 2019. "R&D Spending, Knowledge Capital, and Agricultural Productivity Growth: A Bayesian Approach." *American Journal of Agricultural Economics* 101, no. 1: 291–310. https://doi.org/10.1093/ajae/aay039.

Barbier, E. B., and J. C. Burgess. 2019. "Scarcity and Safe Operating Spaces: The Example of Natural Forests." *Environmental and Resource Economics* 74, no. 3: 1077–1099. https://doi.org/10.1007/s10640-019-00360-9.

Baldos, U., K. Fuglie, A. Cisneros-Pineda, and T. W. Hertel. 2025. "Adoption of improved crop varieties limited biodiversity losses, terrestrial carbon emissions and cropland expansion in the tropics." *Proceedings of the National Academy of Sciences* 122, no. 6: e2404839122. https://doi.org/10.1073/pnas.2404839122.

Barbier, E. B., and J. C. Burgess. 2024. "Economics of Peatlands Conservation, Restoration and Sustainable Management." SSRN Electronic Journal. https://doi.org/10.2139/ssrn.4695533.

Biermann, F., and R. E. Kim. 2020. "The Boundaries of the Planetary Boundary Framework: A Critical Appraisal of Approaches to Define a 'Safe Operating Space' for Humanity." *Annual Review of Environment and Resources* 45, no. 1: 497–521. https://doi.org/10.1146/annurev-environ-012320-080337.

Brainerd, E., and N. Menon. 2014. "Seasonal Effects of Water Quality: The Hidden Costs of the Green Revolution to Infant and Child Health in India." *Journal of Development Economics* 107: 49–64. https://doi.org/10.1016/j.jdeveco.2013.11.004.

Bureau, J. C., and J. Antón. 2022. "Agricultural Total Factor Productivity and the Environment: A Guide to Emerging Best Practices in Measurement." OECD. https://doi.org/10.1787/6fe2f9e0-en.

Busch, J. M., and K. Ferretti-Gallon. 2017. "What Drives Deforestation and What Stops It? A Meta-Analysis." *Review of Environmental Economics and Policy* 11, no. 1: 3–23. https://doi.org/10.1093/reep/rew013.

Ceballos, G., P. R. Ehrlich, A. D. Barnosky, A. García, R. M. Pringle, and T. M. Palmer. 2015. "Accelerated Modern Human–Induced Species Losses: Entering the Sixth Mass Extinction." *Science Advances* 1, no. 5: e1400253. https://doi.org/10.1126/sciadv.1400253.

Chambers, R. G. 2016. "Thinking About Agricultural Productivity Accounting in the Presence of By-Products." White Paper No. 16–02. Department of Agricultural and Resource Economics, Univ, Maryland.

Chaudhary, A., and T. Kastner. 2016. "Land Use Biodiversity Impacts Embodied in International Food Trade." *Global Environmental Change* 38: (May):195–204. https://doi.org/10.1016/j.gloenvcha.2016.03.013.

Christianson, L., J. Tyndall, and M. Helmers. 2013. "Financial Comparison of Seven Nitrate Reduction Strategies for Midwestern Agricultural Drainage." *Water Resources and Economics* 2–3: 30–56. https://doi.org/10.1016/j.wre.2013.09.001.

Cisneros-Pineda, A., A. Chaudhary, U. L. C. Baldos, Y. Sung, and T. W. Hertel. 2024. "Evolving Demographic Trends Will Shape Planetary Biodiversity." Cambridge University, UK. https://sites.google.com/view/xxvbioecon/home.

Coomes, O. T., B. L. Barham, G. K. MacDonald, N. Ramankutty, and J. P. Chavas. 2019. "Leveraging Total Factor Productivity Growth for Sustainable and Resilient Farming." *Nature Sustainability* 2, no. 1: 22–28. https://doi.org/10.1038/s41893-018-0200-3.

Crippa, M., E. Solazzo, D. Guizzardi, F. Monforti-Ferrario, F. N. Tubiello, and A. Leip. 2021. "Food Systems Are Responsible for a Third of Global Anthropogenic GHG Emissions." *Nature Food* 2, no. 3: 198–209. https://doi.org/10.1038/s43016-021-00225-9.

Dangar, S., A. Asoka, and V. Mishra. 2021. "Causes and Implications of Groundwater Depletion in India: a Review." *Journal of Hydrology* 596: 126103. https://doi.org/10.1016/j.jhydrol.2021.126103.

Dinerstein, E., C. Vynne, E. Sala, et al. 2019. "A Global Deal for Nature: Guiding Principles, Milestones, and Targets." *Science Advances* 5, no. 4: eaaw2869. https://doi.org/10.1126/sciadv.aaw2869.

Ehui, S. K., and D. S. C Spencer. 1993. "Measuring the Sustainability and Economic Viability of Tropical Farming Systems: A Model From Sub-Saharan Africa." *Agricultural Economics* 9, no. 4: 279–296. https://doi.org/10.1016/0169-5150(93)90018-8.

Ewert, F., R. Baatz, and R. Finger. 2023. "Agroecology for a Sustainable Agriculture and Food System: From Local Solutions to Large-Scale Adoption." *Annual Review of Resource Economics* 15: 351–381. https://doi.org/10.1146/annurev-resource-102422-090105.

Exposito-Alonso, M., T. R. Booker, L. Czech, et al. 2022. "Genetic Diversity Loss in the Anthropocene." *Science* 377, no. 6613: 1431–1435. https://doi.org/10.1126/science.abn5642.

Famiglietti, J. S. 2014. "The Global Groundwater Crisis." *Nature Climate Change* 4, no. 11: 945–948. https://doi.org/10.1038/nclimate2425.

Fan, M., S. Gabbard, A. A. Pena, and J. M. Perloff. 2015. "Why Do Fewer Agricultural Workers Migrate Now?" *American Journal of Agricultural Economics* 97, no. 3: 665–679. https://doi.org/10.1093/ajae/aau115.

Food and Agriculture Organisation of the United Nations, UN. 2023. "The State of Food and Agriculture." https://doi.org/10.4060/cc7724en.

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Fuglie, K., S. Ray, U. L. C. Baldos, and T. W. Hertel. 2022. "The R&D Cost of Climate Mitigation in Agriculture." *Applied Economic Perspectives and Policy* 44, no. 4: 1955–1974. https://doi.org/10.1002/aepp.13245.

Fuglie, K. O., T. Benton, Y. Sheng, J. Hardelin, K. Mondelaers, and D. Laborde. 2016. "Metrics of Sustainable Agricultural Productivity." G-20 MACS White Paper.

Fuglie, K. O. 2018. "R&Dcapital,R&Dspillovers, and productivity growth in world agriculture." >*Appl. Econ. Persp. Policy* 40, no. 3: 421–444.

Fuglie, K. O., and R. G. Echeverria. 2024. "The Economic Impact of CGIAR-Related Crop Technologies on Agricultural Productivity in Developing Countries, 1961–2020." *World Development* 176: 106523. https://doi.org/10.1016/j.worlddev.2023.106523.

Fuglie, K. O., T. W. Hertel, D. B. Lobell, and N. B. Villoria. 2024. "Agricultural Productivity and Climate Mitigation." *Annual Review of Resource Economics* 16: 21–40.

Galloway, J. N., J. D. Aber, J. Willem Erisman, et al. 2003. "The Nitrogen Cascade." *Bioscience* 53, no. 4: 341–356. https://doi.org/10.1641/0006-3568(2003)053[0341:TNC]2.0.CO;2.

Gautam, M., D. Laborde, A. Mamun, W. Martin, V. Piñeiro, and R. Vos. 2022. "Repurposing Agricultural Policies and Support: Options to Transform Agriculture and Food Systems to Better Serve the Health of People, Economies, and the Planet." World Bank.

Gerten, D., V. Heck, J. Jägermeyr, et al. 2020. "Feeding Ten Billion People Is Possible Within Four Terrestrial Planetary Boundaries." *Nature Sustainability* 3, no. 3: 200–208. https://doi.org/10.1038/s41893-019-0465-1.

Gliessman, S. 2015. "Agroecology: A Growing Field." *Agroecology and Sustainable Food Systems* 39, no. 1: 1–2. https://doi.org/10.1080/21683565.2014.965869.

Golub, A. A., B. B. Henderson, T. W. Hertel, P. J. Gerber, S. K. Rose, and B. L. Sohngen. 2012. "Global Climate Policy Impacts on Livestock, Land Use, Livelihoods, and Food Security." *Proceedings of the National Academy of Sciences* September, 1–6. https://doi.org/10.1073/pnas.1108772109.

Golub, A. A., T. W. Hertel, H. L. Lee, S. K. Rose, and B. L. Sohngen. 2009. "The Opportunity Cost of Land Use and the Global Potential for Greenhouse Gas Mitigation in Agriculture and Forestry." *Resource and Energy Economics* 31, no. 4: 299–319. https://doi.org/10.1016/j.reseneeco.2009.04.007.

Haqiqi, I., L. Bowling, S. Jame, U. L. C. Baldos, J. Liu, and T. W. Hertel. 2023. "Global Drivers of Local Water Stresses and Global Responses to Local Water Policies in the United States." *Environmental Research Letters* 18, no. 6: 065007. https://doi.org/10.1088/1748-9326/acd269.

Haqiqi, I., J. Doelman, T. Hertel, E. Stehfest, and H. Luchtenbelt. 2024. "Implications of the 30×30 Biodiversity Conservation Targets: Quantifying Uncertainty in Local and Global Food System Projections." Presented during the 27th Annual Conference on Global Economic Analysis. Presented during the 27th Annual Conference on Global Economic Analysis (Fort Collins, Colorado, USA). http://www.gtap.agecon.purdue.edu/resources/res_display.asp?RecordID=7323.

Haqiqi, I., and T. W. Hertel. 2024. *SIMPLE-G: a Gridded Economic Approach to Sustainability Analysis of the Earth's Land and Water Resources*. Springer.

Haqiqi, I., C. J. Perry, and T. W. Hertel. 2022. "When the Virtual Water Runs out: Local and Global Responses to Addressing Unsustainable Groundwater Consumption." *Water International* 47, no. 7: 1060–1084. https://doi.org/10.1080/02508060.2023.2131272.

Hasegawa, T., S. Fujimori, P. Havlík, et al. 2018. "Risk of Increased Food Insecurity Under Stringent Global Climate Change Mitigation Policy." *Nature Climate Change* 8, no. 8: 699–703. https://doi.org/10.1038/s41558-018-0230-x.

Havlík, P., H. J. P. Valin, M. Gusti, et al. 2015. "Climate Change Impacts and Mitigation in the Developing World: An Integrated Assessment of the Agriculture and Forestry Sectors." WPS7477. The World Bank. http://documents.worldbank.org/curated/en/866881467997281798/Climate-change-impacts-and-mitigation-in-the-developing-world-an-integrated-assessment-of-the-agriculture-and-forestry-sectors.

Heisey, P. W., and K. O. Fuglie. 2018. "Agricultural Research Investment and Policy Reform in High-Income Countries." ERR-249. U.S. Department of Agriculture, Economic Research Service.

Henderson, B., A. Golub, D. Pambudi, et al. 2017. "The Power and Pain of Market-Based Carbon Policies: A Global Application to Greenhouse Gases From Ruminant Livestock Production." *Mitigation and Adaptation Strategies for Global Change* January: 1–21. https://doi.org/10.1007/s11027-017-9737-0.

Hertel, T. W. 2021. "Growing Food and Protecting Nature Don't Have to Conflict—Here's How They Can Work Together." The Conversation. March 9, 2021. http://theconversation.com/growing-food-and-protecting-nature-dont-have-to-conflict-heres-how-they-can-work-together-146069.

Hertel, T. W., E. Irwin, S. Polasky, and N. Ramankutty. 2023. "Focus on Global–Local–Global Analysis of Sustainability." *Environmental Research Letters* 18, no. 10: 100201. https://doi.org/10.1088/1748-9326/acf8da.

Hertel, T. W., N. Ramankutty, and U. L. C. Baldos. 2014. "Global Market Integration Increases Likelihood That a Future African Green Revolution Could Increase Crop Land Use and CO2 Emissions." *Proceedings of the National Academy of Sciences* 111, no. 38: 13799–13804.

Hertel, T. W., T. A. P. West, J. Börner, and N. B. Villoria. 2019. "A Review of Global-Local-Global Linkages in Economic Land-Use/Cover Change Models." *Environmental Research Letters* 14, no. 5: 053003. https://doi.org/10.1088/1748-9326/ab0d33.

Humpenöder, F., K. Karstens, H. Lotze-Campen, et al. 2020. "Peatland Protection and Restoration Are Key for Climate Change Mitigation." *Environmental Research Letters* 15, no. 10: 104093. https://doi.org/10.1088/1748-9326/abae2a.

Hussein, Z., T. W. Hertel, and A. A. Golub. 2013. "Climate Change Mitigation Policies and Poverty in Developing Countries." *Environmental Research Letters* 8, no. 3: 035009. https://doi.org/10.1088/1748-9326/8/3/035009.

Interagency Working Group, U. S. 2021. "Technical Support Document: Social Cost of Carbon, Methane, and Nitrous Oxide." Technical Report. Washington D.C.

Iowa State University. 2013. *Iowa Nutrient Reduction Strategy*. Iowa Department of Agriculture and Land Stewardship, Iowa Department of Natural Resources, and Iowa State University College of Agriculture and Life Sciences. https://www.nutrientstrategy.iastate.edu/.

IPBES Secretariat. 2019. "Global Assessment Report on Biodiversity and Ecosystem Services." May 17, 2019. https://www.ipbes.net/node/35274.

Jenkins, C. N., S. L. Pimm, and L. N. Joppa. 2013. "Global Patterns of Terrestrial Vertebrate Diversity and Conservation." *Proceedings of the National Academy of Sciences* 110, no. 28: E2602–E2610. https://doi.org/10.1073/pnas.1302251110.

Johnson, J. A., U. L. Baldos, E. Corong, et al. 2023. "Investing in Nature Can Improve Equity and Economic Returns." *Proceedings of the National Academy of Sciences* 120, no. 27: e2220401120. https://doi.org/10.1073/pnas.2220401120.

Leclère, D., M. Obersteiner, M. Barrett, et al. 2020. "Bending the Curve of Terrestrial Biodiversity Needs an Integrated Strategy." *Nature* 585, no. 7826: 551–556. https://doi.org/10.1038/s41586-020-2705-y.

Lenzen, M., D. Moran, K. Kanemoto, B. Foran, L. Lobefaro, and A. Geschke. 2012. "International Trade Drives Biodiversity Threats in Developing Nations." *Nature* 486, no. 7401: 109–112. https://doi.org/10.1038/nature11145.

Liu, J., L. C. Bowling, C. J. Kucharik, et al. 2023. "Tackling Policy Leakage and Targeting Hotspots Could Be Key to Addressing the 'Wicked' Challenge of Nutrient Pollution From Corn Production in the

15740862, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/agec.70008, Wiley Online Library on [30/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

U.S." Environmental Research Letters. https://doi.org/10.1088/1748-9326/acf727.

Liu, J., T. W. Hertel, F. Taheripour, T. Zhu, and C. Ringler. 2014. "International Trade Buffers the Impact of Future Irrigation Shortfalls." *Global Environmental Change* 29: 22–31. https://doi.org/10.1016/j.gloenvcha.2014.07.010.

Lotze-Campen, H., C. Müller, A. Bondeau, S. Rost, A. Popp, and W. Lucht. 2008. "Global Food Demand, Productivity Growth, and the Scarcity of Land and Water Resources: A Spatially Explicit Mathematical Programming Approach." *Agricultural Economics* 39, no. 3: 325–338. https://doi.org/10.1111/j.1574-0862.2008.00336.x.

Martin, W., and R. Vos. 2024. "Options for Reducing Greenhouse Gas Emissions from Agriculture and Food Systems." In Ravello, Italy.

McCarl, B. A., and U. A. Schneider. 2001. "Climate Change: Greenhouse Gas Mitigation in U.S. Agriculture and Forestry." *Science* 294, no. 5551: 2481–2482. https://doi.org/10.1126/science.1064193.

Mehrabi, Z., E. C. Ellis, and N. Ramankutty. 2018. "The Challenge of Feeding the World While Conserving half the Planet." *Nature Sustainability* 1, no. 8: 409–412. https://doi.org/10.1038/s41893-018-0119-8.

Murray, B. C., B. L. Sohngen, A. Sommer, et al. 2005. "Greenhouse Gas Mitigation Potential in US Forestry and Agriculture." EPA Report 430-R-05-006.

Otto, F. E. L. 2017. "Attribution of Weather and Climate Events." *Annual Review of Environment and Resources* 42, no. 1: 627–646. https://doi.org/10.1146/annurev-environ-102016-060847.

Persson, L., B. M. Carney Almroth, C. D. Collins, et al. 2022. "Outside the Safe Operating Space of the Planetary Boundary for Novel Entities." *Environmental Science & Technology* 56, no. 3: 1510–1521. https://doi.org/10.1021/acs.est.1c04158.

Qaim, M., R. Barrangou, and P. C. Ronald. 2024. "Sustainability of Animal-Sourced Foods and Plant-Based Alternatives." *Proceedings of the National Academy of Sciences* 121, no. 50: e2400495121. https://doi.org/10.1073/pnas.2400495121.

Raworth, K. 2017. *Doughnut Economics: Seven Ways to Think like a 21st-Century Economist*. Chelsea Green Publishing.

Ray, S., I. Haqiqi, A. E. Hill, J. E. Taylor, and T. W. Hertel. 2023. "Labor Markets: A Critical Link Between Global-Local Shocks and Their Impact on Agriculture." *Environmental Research Letters* 18, no. 3: 035007. https://doi.org/10.1088/1748-9326/acb1c9.

Resnick Edited by Danielle, and Johan Swinnen eds. 2024. *The Political Economy of Food System Transformation: Pathways to Progress in a Polarized World*. Oxford University Press.

Richardson, K., W. Steffen, W. Lucht, et al. 2023. "Earth Beyond Six of Nine Planetary Boundaries." *Science Advances* 9, no. 37: eadh2458. https://doi.org/10.1126/sciadv.adh2458.

Rockström, J., J. Gupta, D. Qin, et al. 2023. "Safe and Just Earth System Boundaries." *Nature* 619, no. 7968: 102–111. https://doi.org/10.1038/s41586-023-06083-8.

Rockström, J., W. Steffen, K. Noone, et al. 2009. "A Safe Operating Space for Humanity." *Nature* 461, no. 7263: 472–475. https://doi.org/10.1038/461472a.

Seppelt, R., C. Arndt, M. Beckmann, E. A. Martin, and T. W. Hertel. 2020. "Deciphering the Biodiversity–Production Mutualism in the Global Food Security Debate." *Trends in Ecology and Evolution* 35, no. 11: 1011–1020.

Smerald, A., D. Kraus, J. Rahimi, et al. 2023. "A Redistribution of Nitrogen Fertiliser Across Global Croplands Can Help Achieve Food Security Within Environmental Boundaries." *Communications Earth & Environment* 4, no. 1: 1–11. https://doi.org/10.1038/s43247-023-00970-8.

Stevenson, J., D. Byerlee, N. B. Villoria, T. Kelley, and M. Maredia. 2012. "Agricultural Technology, Global Land Use 488 and Deforestation: A Review." *Proceedings of the National Academy of Sciences* In review.

Ürge-Vorsatz, D., S. Tirado-Herrero, N. L. Andrew, and F. Lecocq. 2014. "Measuring the Co-Benefits of Climate Change Mitigation." *Annual Review of Environment and Resources* 39: (October):549–582. https://doi.org/10.1146/annurev-environ-031312-125456.

Valin, H. J. P., P. Havlík, N. Forsell, et al. 2013. "Description of the GLOBIOM (IIASA) Model and Comparison With the MIRAGE-BioF (IFPRI) Model." ENER/C1/428-2012. IIASA, Ecofys and E4tech. http://globiom-iluc.eu/wp-content/uploads/2014/02/Describing-GLOBIOM-and-comparison-with-MIRAGE-BioF_October-2013.pdf.

Valin, H. J. P., T. W. Hertel, B. L. Bodirsky, T. Hasegawa, and E. Stehfest. 2021. "Achieving Zero Hunger by 2030 A Review of Quantitative Assessments of Synergies and Tradeoffs amongst the UN Sustainable Development Goals." Report. Center for Development Research (ZEF) in cooperation with the Scientific Group for the UN Food System Summit 2021. https://doi.org/10.48565/scgr2021-2337.

Villoria, N. B. 2019. "Consequences of Agricultural Total Factor Productivity Growth for the Sustainability of Global Farming: Accounting for Direct and Indirect Land Use Effects." *Environmental Research Letters* 14, no. 12: 125002. https://doi.org/10.1088/1748-9326/ab4f57.

Vitousek, P. M., J. D. Aber, R. W. Howarth, et al. 1997. "Human Alteration of the Global Nitrogen Cycle: Sources and Consequences." *Ecological Applications* 7, no. 3: 737–750. https://doi.org/10.1890/1051-0761(1997)007[0737:HAOTGN]2.0.CO;2.

Wada, Y., and M. F. P. Bierkens. 2014. "Sustainability of Global Water Use: Past Reconstruction and Future Projections." *Environmental Research Letters* 9, no. 10: 104003. https://doi.org/10.1088/1748-9326/9/10/104003.

Wada, Y., L. P. H. Van Beek, and M. F. P. Bierkens. 2012. "Nonsustainable Groundwater Sustaining Irrigation: A Global Assessment." *Water Resources Research* 48, no. 6: 2011WR010562. https://doi.org/10.1029/2011WR010562.

Wada, Y., D. Wisser, and M. F. P. Bierkens. 2014. "Global Modeling of Withdrawal, Allocation and Consumptive Use of Surface Water and Groundwater Resources." *Earth System Dynamics* 5, no. 1: 15–40. https://doi.org/10.5194/esd-5-15-2014.

Wezel, A., M. Casagrande, F. Celette, J. F. Vian, A. Ferrer, and J. Peigné. 2014. "Agroecological Practices for Sustainable Agriculture. A Review." *Agronomy for Sustainable Development* 34, no. 1: 1–20.

Wisser, D., B. M. Fekete, C. J. Vörösmarty, and A. H. Schumann. 2010. "Reconstructing 20th Century Global Hydrography: A Contribution to the Global Terrestrial Network- Hydrology (GTN-H)." *Hydrology and Earth System Sciences* 14, no. 1: 1–24. https://doi.org/10.5194/hess-14-1-2010.

Wuepper, D., I. Wiebecke, L. Meier, et al. 2024. "Agri-Environmental Policies From 1960 to 2022." *Nature Food* 5, no. 4: 323–331. https://doi.org/10.1038/s43016-024-00945-8.

Yu, Z., J. Loisel, D. P. Brosseau, D. W. Beilman, and S. J. Hunt. 2010. "Global Peatland Dynamics Since the Last Glacial Maximum." *Geophysical Research Letters* 37, no. 13. https://doi.org/10.1029/2010GL043584.

Zuidema, S., J. Liu, M. G. Chepeliev, et al. 2023. "US Climate Policy Yields Water Quality Cobenefits in the Mississippi Basin and Gulf of Mexico." *Proceedings of the National Academy of Sciences* 120, no. 43: e2302087120. https://doi.org/10.1073/pnas.2302087120.

Zuidema, S., W. M. Wollheim, C. J. Kucharik, and R. B. Lammers. 2024. "Existing Wetland Conservation Programs Miss Nutrient Reduction Targets." *PNAS Nexus* 3, no. 4: pgae129. https://doi.org/10.1093/pnasnexus/pgae129.

15740862, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/agec.70008, Wiley Online Library on [30/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License